THE ERA OF EXREME WEATHER

Graphic by **NILANJAN DAS**

1.5°C
Global rise in annual average temperature since pre-industrial (1850–1900) levels

GLOBAL WARMING

- Hottest year on record globally: 2024
- India's annual mean temperature has risen by 0.69°C since 2001
- For a tropical country, India already has a high base average temperature of over 25°C
- Moreover, the Himalayan region has seen a 2.5°C rise over pre-industrial levels
- The country's 10 warmest years in 120 years have all been in the past 15 years

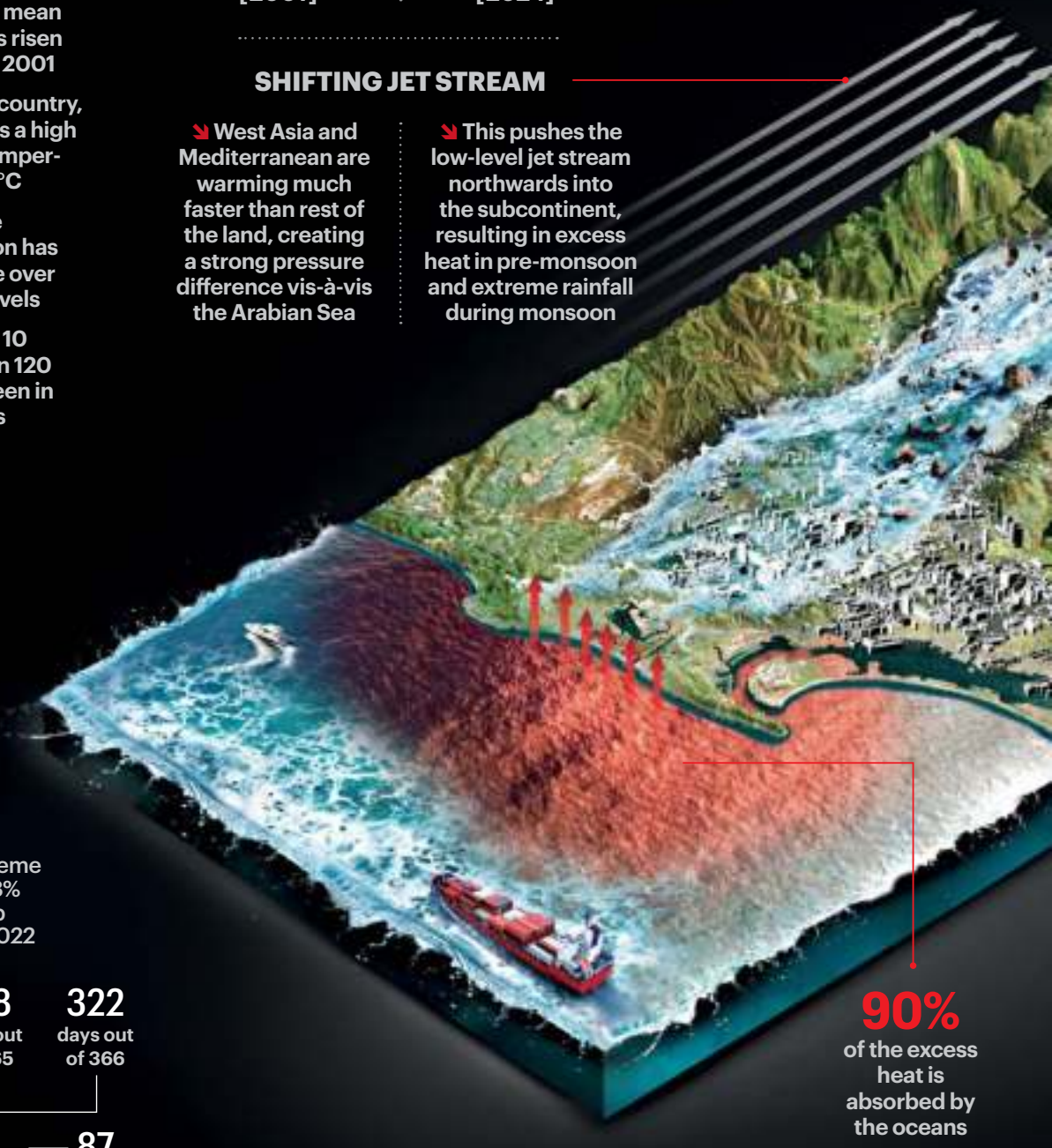
ANNUAL MEAN TEMPERATURE IN INDIA

25.05°C [2001]

25.74°C [2024]

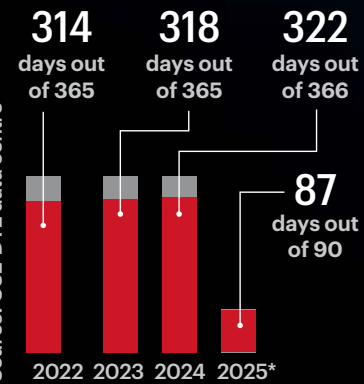
SHIFTING JET STREAM

- West Asia and Mediterranean are warming much faster than rest of the land, creating a strong pressure difference vis-à-vis the Arabian Sea
- This pushes the low-level jet stream northwards into the subcontinent, resulting in excess heat in pre-monsoon and extreme rainfall during monsoon



EXTREME WEATHER EVENTS

In 2024, India recorded extreme weather on 88% of the days, up from 86% in 2022



*January to March
Source: CSE-DTE data centre

90%
of the excess heat is absorbed by the oceans

WEAKENING OCEAN CURRENTS

- Global warming weakens the Atlantic Ocean currents by adding fresh water from melting polar ice, which slows the northward flow of warm water
- This disruption can shift tropical rain belts and weaken the Indian monsoon, intensifying heat waves and droughts across South Asia